

$$\frac{18}{19} \Rightarrow 95\%$$

0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12	12.5	13	13.5	14	14.5	15	15.5	16	16.5	17	17.5	18	18.5	19	19.5	20
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Name: _____

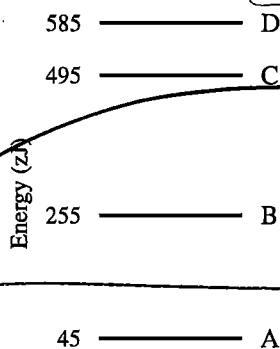
These questions assess your understanding of the material from Chapters 5, 6 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Several types of electromagnetic radiation (EMR) are: microwaves, radio waves, orange light, and ultraviolet light. Order these types of EMR from lowest frequency to highest frequency.

ANSWER: radio < micro < orange < UV

2. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the longest wavelength? You must explain your answer choice for full credit.

the emission transition is size inversely proportional to the wavelength so the shortest transition is the longest wavelength, emission is going down in energy so 585 $\rightarrow$ 495 is the longest λ

* what is emitted? PHOTON"lowest ΔE "
not "shortest""longest" is correct $\because \lambda$ is a lengthANSWER: D $\rightarrow$ C

3. Fill in the information missing from the following table. For each subshell, you must provide the principal quantum number n , the angular momentum quantum number l , and the maximum number of electrons that can fit in the subshell N_e . Credit is awarded only if the table is completed correctly. You may fill in the table directly; there is no need to redraw it.

Subshell	n	l	N_e
3d	3	2	10
5s	5	0	2
4s	4	0	2

ANSWER: _____

4. Suppose a popular FM-radio station broadcasts radio waves with a frequency of 93.21 MHz. Calculate the wavelength of these radio waves.

ANSWER: 3.216m

$$\lambda = c/\nu$$

$$\nu = 93.21 \text{ MHz} = 93210000$$

$$\lambda = \frac{(2.998 \times 10^8)}{93210000} = 3.216 \text{ m}$$

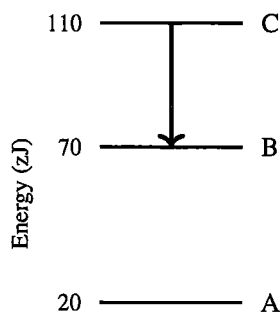
5. A particular cookie has an energy content of 120 kcal. What is this in kilojoules?

ANSWER: 50. $\times 10^3$ KJ

$$120 \text{ kcal} \cdot \frac{4.184 \text{ KJ}}{1 \text{ kcal}} = 502.08 = 50. \times 10^3$$

+ 5

6. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from C to B, as shown by the arrow. Calculate the wavelength of the photon that would be absorbed or emitted. Report your answer to three significant figures. Note that the SI prefix zepto (z) means 10^{-21} .



$$E = hc/\lambda \rightarrow \lambda = hc/E$$

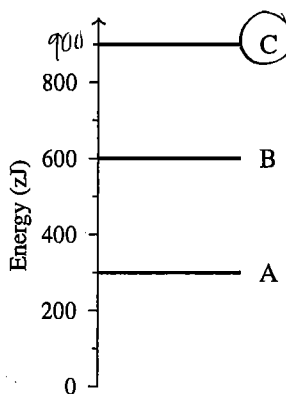
$$E = 110 - 70 = 40 \times 10^{-21} \text{ J}$$

$$\lambda = \frac{(6.626 \times 10^{-34})(2.998 \times 10^8)}{40 \times 10^{-21}}$$

$$\lambda = 4.97 \times 10^{-6} \text{ m}$$

ANSWER: $4.97 \times 10^{-6} \text{ m}$

7. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its second excited state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: 900 zJ

8. Calculate the wavelength of the line in the emission line spectrum of hydrogen caused by the transition of the electron from an orbital with $n = 8$ to an orbital with $n = 2$. Round your answer to 3 significant figures.

$$1/\lambda = R_{\infty} \cdot (1/n_1^2 - 1/n_2^2) \quad n_1 = 2, n_2 = 8$$

$$1/\lambda = (1.097 \times 10^7) (1/4 - 1/64) = 1/2571093.75 = 3.89 \times 10^{-7} \text{ m}$$

ANSWER: $3.89 \times 10^{-7} \text{ m}$

9. A mixture of methane and carbon dioxide gas is expanded from a volume of 62.1 L to a volume of 75.5 L while the pressure is held constant at 86 atm. Calculate the work done on the gas mixture. Report your answer in SI unit of work and to 3 significant figures.

$$W = -p \cdot \Delta V$$

$$W = -(86)(101.325) \cdot (75.5 - 62.1) = -116766.93$$

$$= -1.17 \times 10^5 \text{ J}$$

ANSWER: $-1.17 \times 10^5 \text{ J}$

+4

10. A chemist carefully measures the amount of heat needed to raise the temperature of a 1.1-kg sample of a pure substance from 2.26°C to 21.°C. The experiment shows that 4.02 kJ of heat is needed. What can the chemist report as the value of specific heat capacity of the substance?

$$q = mc\Delta T \rightarrow c = q/m\Delta T$$

$$c = \frac{4.02 \text{ kJ}}{(1.1 \text{ kg})(21 - 2.26)} = 0.195013 = \boxed{0.20 \text{ kJ/kg}^\circ\text{C}}$$

ANSWER: 0.20 kJ/kg°C

11. Imagine an alternate universe in which the value of the Planck constant is $1.44 \times 10^{-31} \text{ Js}$. In that universe, is quantum mechanics necessary to accurately describe a $5.0 \times 10^{-13} \text{ m}$ -wide object that has a mass of $2.2 \times 10^{-19} \text{ kg}$ and is moving at $6.5 \times 10^{-5} \text{ m/s}$? Write "yes" if quantum mechanics is necessary and "no" if it is not, then explain why you made your choice. Answers lacking supporting explanation will earn no credit.

$$\lambda = h/p \quad p = mv \rightarrow \lambda = h/mv$$

$$\lambda = \frac{(1.44 \times 10^{-31})}{(2.2 \times 10^{-19})(6.5 \times 10^{-5})} = 1.0 \times 10^{-8} \text{ m} > 5 \times 10^{-13} \text{ m}$$

yes q.m. is needed b/c the deBroglie λ is larger than the width

ANSWER: _____

12. Grace puts a 0.30-kg aluminum pot on the stove and fills it with 0.76 liters of water. Both the pot and the water are initially at 6.19°C. How much heat is required to bring them both to 59.°C? The specific heat value of water is 4186 J/kg°C and for aluminum it is 900 J/kg°C. Recall the density of water is 1.000 g/cc.

$$q = mc\Delta T \quad 59 - 6.19 = 52.81^\circ\text{C}$$

ANSWER: $1.8 \times 10^5 \text{ J}$

$$q_{\text{pot}} = (0.30)(900)(52.81^\circ\text{C}) = 14258.7$$

$$q_{\text{w}} = (0.76)(4186)(52.81) = 168007.62$$

$$= \boxed{1.8 \times 10^5 \text{ J}}$$

13. An atom of a particular element has a radius of 109 pm and the average orbital speed of the electrons in it is approximately $1.09 \times 10^7 \text{ m/s}$. Calculate the least possible uncertainty in a measurement of the speed of an electron in this atom. Write your answer as a percentage of the average speed and round it to two significant figures.

$$109 \text{ pm} = 1.09 \times 10^{-10} \text{ m}$$

$$\Delta x \cdot \Delta p \geq h/4\pi \quad (p = m \cdot \Delta v)$$

$$\Delta x \cdot m \Delta v \geq h/4\pi$$

$$\Delta v = \frac{h/4\pi \cdot m \cdot \Delta x}{m \cdot \Delta x} = \frac{(6.626 \times 10^{-34})}{4\pi(9.11 \times 10^{-31})(1.09 \times 10^{-10})} = \frac{531002.6551}{1.09 \times 10^7} = 0.0487 \cdot 100 = \boxed{4.9\%}$$

ANSWER: 4.9%

14. Several types of electromagnetic radiation (EMR) are: microwaves, radio waves, gamma rays, and red light. Order these types of EMR from lowest energy to highest energy.

ANSWER: Radio < micro < red < gamma

15. Several types of electromagnetic radiation (EMR) are: infrared light, microwaves, yellow light, and green light. Order these types of EMR from shortest wavelength to longest wavelength.

ANSWER: green < yellow < IR < micro

16. Suppose that there is heat transfer of 34 J to a system, while the system does 55 J of work. Later, there is heat transfer of 43 J out of the system while 58 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: -6 J

$$34 \text{ J} - 55 \text{ J} = -21$$

$$-43 \text{ J} + 58 \text{ J} = +15$$

$$\boxed{-6 \text{ J}}$$

+7

17. List the following electron subshells in order of increasing energy. Assume these subshells are in an atom with many electrons.

~~2p, 4p, 4d, 3d, 3p~~

2p 3p 3d 4p 4d

ANSWER: 2p 3p 3d 4p 4d ✓

18. Measurements show that the enthalpy of a mixture of gaseous reactants has decreased by 265.2 kJ during a certain chemical reaction, which was carried out at a constant pressure. Evidenced by the volume change, -174 kJ of work must have been done on the mixture during the reaction. Calculate the change in energy of the gas mixture during the reaction and report whether the reaction is endothermic, exothermic, or neither. You must explain your answer choice for full credit.

~~$\Delta E = Q + W \rightarrow \Delta E = \Delta H + W$~~

$\Delta E = Q + W$

$\Delta H = Q$ if $\Delta P = 0$

ANSWER: 4.61×10^4 endothermic

$\Delta E = (-265.2) + (-174) = -439.2$ (3 SF) $\Delta E = 4.61 \times 10^4$ endothermic b/c the ΔE is greater than 0 & positive

correct: -439 kJ

19. It takes 160. kJ/mol to break a bond between two particular elements. Calculate the maximum wavelength of light that will break one of these bonds.

(3 SF)

$\lambda = hc / \Delta E$

ANSWER: $7.48 \times 10^{-7} \text{ m}$ ✓

$160 \text{ kJ/mol} = \frac{160000 \text{ J/mol}}{6.022 \times 10^{23} \text{ mol}^{-1}} = 2.6569 \times 10^{-19}$

$\lambda = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})(2.998 \times 10^8 \text{ m/s})}{2.6569 \times 10^{-19} \text{ J}} = 7.48 \times 10^{-7} \text{ m}$



1. (lowest frequency) radio waves < microwaves < orange light < ultraviolet light (highest frequency)
2. $D \rightarrow C$ $\therefore$ this transition corresponds to the smallest energy difference, resulting in emission of a photon with the longest wavelength and lowest frequency.
3. 3d: $n = 3, l = 2, N_e = 10$... 5s: $n = 5, l = 0, N_e = 2$... 4s: $n = 4, l = 0, N_e = 2$
4. 3.216 m
5. 5.0×10^2 kJ
6. 4.97×10^{-6} m $\therefore$ the energy of the emitted photon is equal to the difference in energy levels and $E = hc/\lambda$
7. 900 zJ $\therefore$ "C" is the level with the third-lowest energy, AKA the second excited state.
8. 3.89×10^{-7} m $\therefore \Delta E = \left(\frac{-R_y}{n_2^2} \right) - \left(\frac{-R_y}{n_1^2} \right)$ and $\Delta E = \frac{hc}{\lambda}$
9. -1.17×10^5 J
10. 0.20 J/g/K
11. Yes $\therefore$ the de Broglie wavelength $\lambda = \frac{h}{mv}$ is larger than the width of the object.
12. 180 kJ
13. 4.9% $\therefore \Delta x \Delta p \geq \frac{h}{4\pi}$ and $\Delta p = m\Delta v$
14. (lowest energy) radio waves < microwaves < red light < gamma rays (highest energy)
15. (shortest wavelength) green light < yellow light < infrared light < microwaves (longest wavelength)
16. -6 J
17. 2p, 3p, 3d, 4p, 4d
18. -439 kJ. Since $\Delta H < 0$, the reaction is exothermic.
19. 7.48×10^{-7} m